Answers and talk notes

Represent Rational Numbers and Their Opposites on the Number Line

What your student is learning

Tonight we find opposites of fractions and decimals — not just whole numbers — and locate both on a number line.

Answers

Ask your student to explain before you look.

1. **C.** −3/4. The opposite of 3/4 is −3/4. The fraction keeps its size and only changes side.
2. **A.** 5.6. The opposite of −5.6 is 5.6.
3. **D.** 1.2 and −1.2. 1.2 and −1.2 are the same distance from 0 on opposite sides.
4. **C.** 2.5 units. An opposite is the same distance from 0, so it is 2.5 units away.
5. **B.** −6.35. The first opposite is 6.35, and the opposite of that is −6.35.
6. **A.** −1.5. The point is 1.5, so its opposite is −1.5.

Talk about it

Is the opposite of a fraction or decimal the same distance from 0? Which side is it on?

Home mission

Put a fork on the table as zero. Put a coin one hand-width to the right of the fork. Ask your student: where does the opposite coin go? Then ask: where does the opposite of the opposite go?

When they are stuck, ask

- Which two whole numbers does this value fall between?
- Same distance from zero — which side now?
- What happens if you take the opposite twice?

Try not to say

- For −3/4, go to −3 and then count 4 more.
- Only whole numbers have opposites.
- The opposite is just the number with a minus added.